Ghislain THEUBO

Paris, France | +33 6 82 86 62 79 | bergerghislaintheubo@gmail.com
LinkedIn | Bilingual English/French | GitHub

Professional Summary

Software Engineer (Ensimag) with strong **Java** foundations, experience on large-scale platforms at **Dassault Systèmes (3DEXPERIENCE)**, and a genuine interest in **language tooling**, **semantic analysis**, and **low-latency APIs**. Comfortable with parsing, indexing, caching, and API design. Built personal projects around a **mini Language Server Protocol (LSP)** stack and a **DSL semantic model** — directly aligned with SysML v2 tooling needs (completion, diagnostics, references, incremental analysis). Eager to contribute to CATIA R&D on an AI-native systems-engineering platform.

Core Strengths

Java (OOP, concurrency) · **Language tooling / LSP notions** · **Parsing & semantic models** · **Indexing / caching** · **REST APIs** · **Performance & incremental analysis** · **Rust (notions)** · **Web (JS/TS)** · **Unit testing & CI** · **English**

Experience

Dassault Systèmes – 3DEXPERIENCE R&D — Vélizy-Villacoublay, France
Software Engineer (Platform Architecture & Performance) — *Apr 2025 – Oct 2025*

- Contributed to the **3DEXPERIENCE Platform** in a 20-person R&D team — exposure to large codebases, quality bars, and multi-team delivery.
- Designed and optimized **low-latency service endpoints**; delivered a **35% performance improvement** via algorithmic work, caching strategies, and data-access tuning.
- Implemented **unit/integration tests** (85% coverage) and CI quality gates; practiced rigorous code review and technical documentation.
- Collaborated with product and senior engineers on **API contracts** and maintainable software design for platform features.

Gustave Roussy Cancer Institute & CentraleSupélec MICS Lab — Paris, France
Backend Engineer (CDD – APIs & Production Systems) — *Jan 2026 – Present*

- Designed backend services (Python/FastAPI, PostgreSQL) with validation, access control, structured logging, and **API-first** design.
- Optimized SQL/indexing (**30%** faster responses); containerized services (Docker) and automated delivery (GitHub Actions).

INRIA Grenoble – Humanlab / Froll'N'Roll — Montbonnot-Saint-Martin, France
Research Intern (Real-Time Systems & C/C++) — *May 2024 – Aug 2024*

- Built real-time modules in **C/FreeRTOS** and Python under strict latency constraints; improved pipeline responsiveness by **40%**.
- Applied solid **data structures** and protocol redesign; documented design choices for open-source collaboration.

Laboratoire d'Informatique de Grenoble (LIG) — Grenoble, France
Cloud Infrastructure Intern — *Feb 2025 – Mar 2025*

- Automated VM migration experiments on OpenStack/Grid'5000; analyzed performance and availability trade-offs.

Personal Projects – Language Tooling, Parsing & Indexing

Mini LSP Server for a Systems DSL | Java, LSP4J, ANTLR, JUnit — GitHub

- Implemented a Language Server providing **completion**, **hover**, **go-to-definition**, **find references**, **rename**, and **diagnostics** — matching IDE services expected for SysML-style tooling.
- Built a **parser + semantic model** (name resolution, symbol table, cross-file workspace) with **incremental re-analysis** on file change.
- Added an in-memory **symbol index** and cache invalidation strategy to keep navigation APIs responsive on multi-file workspaces.

Incremental Indexer & Semantic Query API | Java, Rust (CLI), REST — GitHub

- Designed a small **indexing service** that stores symbols/references and exposes **low-latency semantic queries** over HTTP.
- Prototyped a **Rust** sidecar for hot-path indexing/search (ownership model, no GC pauses) while keeping business logic in Java — exploring polyglot stacks similar to modern IDE tooling.

C++ Polyhedral Mesh Library (KD-Tree & Graphs) | C++, Templates, Catch2 — GitHub

- Header-only library for 3D mesh queries; KD-Tree nearest-neighbor and graph shortest paths — strong algorithmic foundation transferable to large model navigation.

Technical Skills

Languages: **Java** (primary), Python, C/C++, Rust (notions), JavaScript/TypeScript, SQL, Bash
Language tooling: LSP concepts, parsing (ANTLR), symbol tables, indexing, caching, incremental analysis
APIs & Web: REST, JSON, low-latency service design, basic front-end integration
Quality: JUnit, Catch2, CI/CD (GitHub Actions), code reviews, performance profiling
Platforms: Linux, Docker, Git, CMake; prior exposure to **3DEXPERIENCE** R&D practices

Education

Grenoble INP - Ensimag — Grenoble, France
Engineering Degree in Computer Science & Software Engineering — *2023 – 2025*

École Nationale Supérieure Polytechnique de Yaoundé — Yaoundé, Cameroon
Master's Degree in Computer Engineering (Top 12%) — *2019 – 2023*

Languages

English: Bilingual (professional working proficiency)
French: Fluent
Spanish: Intermediate